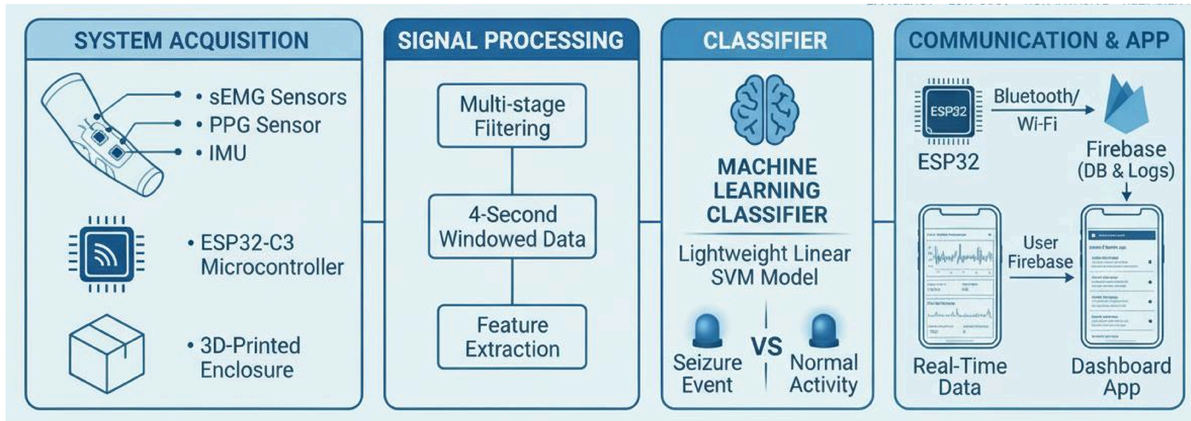
EPILEPSY DETECTION SYSTEM USING WEARABLE SENSORS

ABSTRACT

This project features a low-cost, non-invasive wearable sleeve for real-time epileptic seizure detection. Integrating sEMG, PPG, and IMU sensors with an ESP32-C3, the system utilizes a machine learning pipeline and SVM classifier to identify seizure events. A Flutter-based mobile app provides real-time visualization and event logging via Firebase. The system offers a sustainable, clinically meaningful monitoring solution that operates reliably in real-world environments without requiring centralized infrastructure.

FIGURE



OBJECTIVES

- Develop a non-invasive, low-cost hardware solution
- Implement a real-time signal processing pipeline
- Create a lightweight seizure classification model
- Establish a cloud-based communication and monitoring ecosystem

SUPERVISORS

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SDGs

SDG - 3:
GOOD HEALTH
AND WELL BEING

TOOLS USED

ARDUINO, kaggle, SOLIDWORKS, Flask

GROUP MEMBERS

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